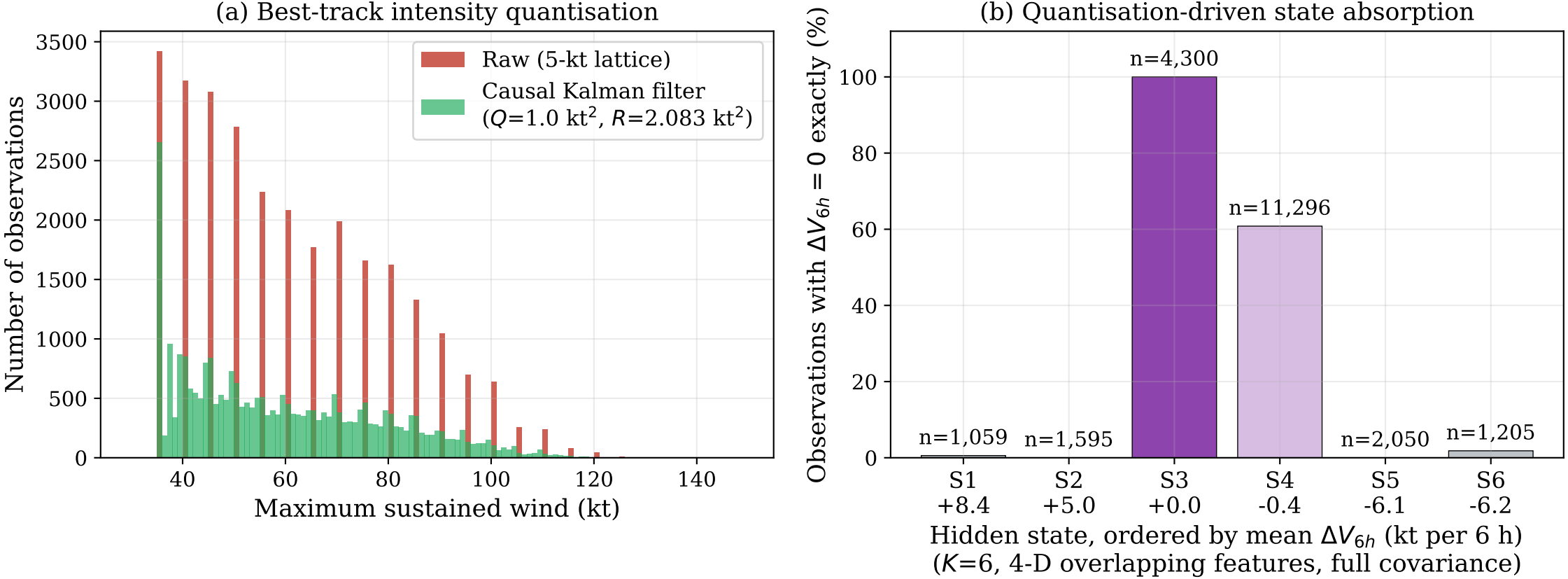
(a) Best-track intensity quantisation
Raw (5-kt lattice)
Causal Kalman filter
(Q=1.0 kt^2, R=2.083 kt^2)
Number of observations
Maximum sustained wind (kt)
0
500
1000
1500
2000
2500
3000
3500
40
60
80
100
120
140
(b) Quantisation-driven state absorption
Observations with $\Delta V_{6h} = 0$ exactly (%)
0
20
40
60
80
100
n=1,059
n=1,595
n=4,300
n=11,296
n=2,050
n=1,205
S1
+8.4
S2
+5.0
S3
+0.0
S4
-0.4
S5
-6.1
S6
-6.2
Hidden state, ordered by mean ΔV_{6h} (kt per 6 h)
(K=6, 4-D overlapping features, full covariance)